
CS 669: Pattern Recognition

Course Code	CS 669
Course Name	Pattern Recognition
L-T-P-C	3-1-0-4
Prerequisites	IC
Intended for	MS/Mtech/PhD/3rd and 4th year B Tech
Distribution	Elective
One-time approval	9th BoA (Oct 2015); 9th Senate (May 2016), in the M.Sc Applied Mathematics curriculum

Note: Credits changed from 3-0-0-3 to 3-1-0-4 in 10th Senate

Course Contents

- **Basics of Probability, Random Processes and Linear Algebra** (recap) [8 Lectures]
 - Probability: independence of events, conditional and joint probability, Bayes theorem
 - Random Processes: Stationary and nonstationary processes, Expectation, Autocorrelation, Cross-Correlation, spectra.
 - Linear Algebra: Inner product, outer product, inverses, eigen values, eigen vectors, singular values, singular vectors.
 - Programming Assignment 1
- **2. Bayes Decision Theory** [8 Lectures]
 - Minimum-error-rate classification
 - Classifiers, Discriminant functions, Decision surfaces
 - Normal density and discriminant functions
 - Discrete features
- **3. Parameter Estimation Methods** [12 Lectures]
 - Maximum-Likelihood estimation: Gaussian case
 - Maximum a Posteriori estimation
 - Bayesian estimation: Gaussian case
 - Unsupervised learning and clustering
 - * Criterion functions for clustering
 - * Algorithms for clustering: K-Means, Hierarchical and other methods
 - * Cluster validation
 - Gaussian mixture models
 - Expectation-Maximization method for parameter estimation
 - Maximum entropy estimation
 - Programming Assignment 2
- **Sequential Pattern Recognition** [4 Lectures]
 - Discrete Time Warping (DTW)
 - Hidden Markov Models (HMMs)
 - Discrete HMMs
 - Continuous HMMs
 - Programming Assignment 3

- **Nonparametric techniques for density estimation** [4 Lectures]
 - Parzen-window method
 - K-Nearest Neighbour method
- **Dimensionality reduction** [4 Lectures]
 - Principal component analysis its relationship to eigen analysis
 - Fisher discriminant analysis Generalised eigen analysis
 - Eigen vectors/Singular vectors as dictionaries.
- **Linear discriminant functions** [8 Lectures]
 - Gradient descent procedures
 - Perceptron
 - Support vector machines
- **Non-metric methods for pattern classification** [4 Lectures]
 - Non-numeric data or nominal data
 - Decision trees: Classification and Regression Trees (CART).
 - Programming Assignment 4/Project

Text Books

1. R.O.Duda, P.E.Hart and D.G.Stork, **Pattern Classification**, John Wiley, 2001
2. S.Theodoridis and K.Koutroumbas, **Pattern Recognition**, 4th Edition, Academic Press, 2009.
3. C.M.Bishop, **Pattern Recognition and Machine Learning**, Springer, 2006.

References

1. Some relevant papers/notes will be put up on the website from time-to-time.
2. Simon Haykin, **Neural Networks: A Comprehensive foundation to Neural Networks or Neural Net-works and Learning Machines**, any edition will do.